set 2 Answer key

Q1 a) Truth table

P	Q	$(P \to Q)$	$(Q \wedge (P \to Q))$	$(Q \wedge (P \to Q)) \to P$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	T

$((P \to Q) \wedge (Q \to R)) \to (P \to R)$

P	Q	R	① $P \to Q$	② $Q \to R$	① ∧ ②	③ $P \to R$	(① ∧ ②) → ③
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	F	T	T
T	F	F	F	T	F	F	T
F	T	T	T	T	T	T	T
F	T	F	T	F	F	F	T
F	F	T	T	T	T	T	F
F	F	F	T	T	T	F	F

its not tautology

Q1 b) PDNF

1) $P \wedge (P \to Q)$

$P \wedge (\neg P \vee Q)$ implication law

$(P \wedge \neg P) \vee (P \wedge Q)$ distributive law

$F \vee (P \wedge Q)$ complement law

$P \wedge Q$ identity law

$F \vee x = x$ PDNF = $P \wedge Q$

Q 1 b) $P \wedge Q$

2) $P \wedge (Q \vee \neg Q) \wedge Q \wedge (P \vee \neg P)$

$(P \wedge Q) \vee (P \wedge \neg Q) \wedge (Q \wedge P) \vee (Q \wedge \neg P)$

$(P \wedge Q) \vee (P \wedge \neg Q) \vee (Q \wedge \neg P)$ - PDNF

Q 2 a) $P + Q = \{5, 8\}$

$P - Q = \{5, 8\}$

$Q - P = \{\}$

$P + Q = \{5, 8\}$

$Q - R = \{\}$

$R - Q = \{8, 9\}$

$Q + R = \{8, 9\}$

$(P + Q) + (Q + R) = \{5, 8\} + \{8, 9\}$

$= \{5, 8, 9\}$

b) $R = \{(1,2), (2,4), (3,6), (4,8), (5,10), (6,12), (7,14), (8,16), (9,18), (10,20)\}$

$S = \{(1,3), (2,6), (3,9), (4,12), (5,15), (6,18), (7,21), (8,24), (9,27), (10,30)\}$

$ROS = \{(1,6),(2,12),(3,18),(4,24),(5,30)\}$

$ROR = \{(1,4),(2,8),(3,12),(4,16),(5,20)\}$

$R\circ(ROR) = \{(1,8),(2,16)\}$

$(ROS)\circ R = \{(1,12)\}$

a) Given $(z,+)$

Subset : $2z = \{\ldots -4,-2,0,2,4\}$

1. identity $0 \in 2z$

2. closure : if $a = 2m$, $b = 2n \in 2z$ then

$a+b = 2m+2n = 2(m+n) \in 2z$

3. Inverse : if $a = 2m \in 2z$ then $-a = -2m \in 2z$

therefore $2z$ is a subgroup of $(z,+)$

let $a,b \in S \longrightarrow a \circ b = a \in S$

$\circ$ is binary operation in S let $a,b,c \in S$

$a \circ (b \circ c) = a \circ b = a$

$(a \circ b) \circ c = a \circ c = a$

$\circ$ is associative in S so

$(S,\circ)$ is a semigroup.

ADCET

Q 3 b) $A = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$ $P = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$

$A^T = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$ $A \cdot A^T = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$

$= \begin{bmatrix} 2 & 1 & 2 & 1 \\ 1 & 2 & 1 & 0 \\ 2 & 1 & 2 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$

$A^T \cdot A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 2 & 1 & 3 \end{bmatrix}$

$A^2 = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 2 & 0 & \\ 0 & 1 & 1 & \\ 0 & 0 & 1 & \end{bmatrix}$

ADCET

$$A^3 = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 1 & 2 \\ 0 & 1 & 2 & 2 \\ 0 & 2 & 1 & 2 \\ 0 & 2 & 0 & 1 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 0 & 2 & 1 & 2 \\ 0 & 1 & 2 & 2 \\ 0 & 2 & 1 & 2 \\ 0 & 2 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 3 & 2 & 3 \\ 0 & 4 & 1 & 3 \\ 0 & 3 & 2 & 3 \\ 0 & 1 & 2 & 2 \end{bmatrix}$$

3 c)

Node #	Node data	No of edge
1	L	2
2	M	0
3	N	1
4	O	1
5	P	1

ptr to edge list

